A Thesis for the Degree of Master of Arts in Economics

Forecasting Volatilities of MSE Stock Returns using Asymmetric GARCH models with fat tails

두터운 분포꼬리가 있는 비대칭 GARCH 모형을 사용하여 MSE 주식 수익률의 변동성 예측

By
Jargalsaikhan Naranchimeg

Department of Economics
Graduate School
GYEONGSANG NATIONAL UNIVERSITY
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Forecasting Volatilities of MSE Stock Returns using Asymmetric GARCH models with fat tails

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By Jargalsaikhan Naranchimeg

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Prof. Eunhee Lee, Thesis Supervisor

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Thesis Committee ______________________________

Chairperson

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(Name and Signature)

Date: 2020.01.16

Department of Economics

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GYEONGSANG NATIONAL UNIVERSITY

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Jargalsaikhan Naranchimeg

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Thesis Supervisor: Prof. Eunhee Lee

Abstract

It is well known that the relationship between return and volatility in equity markets is asymmetric; a negative return is associated with higher volatility than a positive return. In the literature of autoregressive conditional heteroscedasticity (ARCH) type volatility models, various models have been proposed to accommodate this stylized fact.

As another conventional approach that financial return series generally exhibit non-normal characteristics while the typical GARCH model assumes a normal error distribution (Goksan, 2000). Consequently, the typical GARCH model cannot well capture the stylized facts of return series such as heavy tails, excess kurtosis and skewness.

This paper investigates three generalized autoregressive conditional heteroscedasticity (GARCH) models with different distribution of the errors and the leverage effect using data such as 2 major stock indexes of Standard and Poor's 500 daily index, FTSE100 daily index and MSE (Mongolian stock exchange) daily index.

The GARCH, Glosten Jagannathan and Runkle (GJR) version of GARCH (GJR-GARCH) and Exponential GARCH methodology are employed to investigate the dynamics of stock return volatility and the leverage effect in these 3 stock markets using Student-t and normal distribution densities. The study further examines the forecasting performance of each GARCH model using the alternative distribution of the errors.

Based on the results of the estimation for various version GARCH type models, we mainly capture the forecasting performance. Finally, after comparing the Root Mean Square Error (RMSE), choose the best model to predict the conditional variance of the return series on three stock markets.

According to the result of forecasting, asymmetric GARCH model (EGARCH, GJR-GARCH) with student-t distribution has better performance for S&P500, FTSE100 and MSE stock return.

Abstract

금융 또는 경제시계열의 변동성의 움직임과 관련하여 널리 알려진 정형화된 현상 중의 하나는 변동성의 비대칭성 (asymmetric volatility)이다. 이 현상은 음의 수익률은 양의 수익률보다 더 높은 수준의 변동성을 초래하는 것을 말한다. 또 다른 금융 자산수익률의 특징은 비정규성을 갖는다는 것이다. 금융 또는 경제시계열의 변동성을 설명하기 위해 GARCH (Generalized Autoregressive Conditional heteroscedasticity) 모형이 널리 이용되어 왔다. GARCH 모형은 자산수익률의 변동성 밀집현상 (volatility clustering) 등의 특징을 잘 포착해내는 장점을 가진다. 그러나 GARCH 모형에서 조건부 분산은 단지 과거 오차항들의 크기에만 의존할 뿐, 그 부호와는 무관하므로 레버리지효과 (leverage effect)를 반영할 수 없다는 단점을 가진다. 이러한 문제를 해결하기 위해서는 오차항의 부호에 따른 비대칭적 효과를 GARCH 모형에 반영할 필요가 있다. 또한 자산수익률의 두터운 분포꼬리 (fat tail) 특성을 설명하기 위해 정규성 가정은 수정될 필요가 있다.

따라서 이 논문은 S&P500, FTSE100, MSE (Mongolian Stock Exchange) 주가 수익률을 이용하여 다양한 오차 분포의 가정하에서 세 가지 GARCH 모형을 추정하고, 각 시장의 주가 수익률을 예측하였다.

우리의 예측 결과에 따르면 t-분포를 갖는 비대칭 GARCH 모형 (EGARCH, GJR-GARCH)의 경우 S&P 500, FTSE100 와 MSE 주식 수익률에 대해 더 나은 예측 결과를 제공한다.

1. Introduction

Volatility was always considered to be a key variable for assessing the condition of the financial markets and for making decisions by investors, speculators, investment managers and financial regulators. In the actual context of the years 2007-2012, characterized by a continuing financial and economic crisis, terms such as volatility forecast and risk management are very often mentioned by the scientific community and experienced top practitioners all around the world. With time, researchers proposed many and diverse models to forecast volatility, ranging from time series based volatility models (exponential smoothening, Garman-Klass, Autoregressive Conditional Heteroscedasticity etc.)

Autoregressive Conditional Heteroskedasticity (ARCH) model, introduced by Engle (1982) and generalized by Bollerslev (1986) in GARCH model and its various extensions are used to forecast volatility in stock return series (Bollerslev, Chou an d Kroner (1992), Bollerslev, Engle and Nelson (1994) and Kroner and Ng (1998)). The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is the successful model to use for volatility modeling and forecasting financial returns.

It is well known that the relationship between return and volatility in equity markets is asymmetric; a negative return is associated with higher volatility than a positive return. In the literature of autoregressive conditional heteroscedasticity (ARCH) type volatility models, various models have been proposed to accommodate this stylized fact. Theoretical grounding of stock return volatility dates back to Black (1976) theory, which states that stock return volatility is mainly caused by leverage effects. In other words, a decline in a firm's equity holding other things are constant, increases the debt/equity ratio of the firm, thus, inherently increasing its riskiness. Lack of conclusiveness in stock market returns has led to the founding of a number of models measuring leverage effects such as the GARCH. Moreover, some scholars have also developed variant forms of the GARCH model. For example, Nelson (1991) proposed the exponential GARCH (EGARCH) model, and Engle and Bollerslev (1986) proposed the Integrated GARCH (IGARCH) model. Ding, Ganger, and Engle (1993) first mentioned the Power GARCH (PGARCH) model, while

Glosten, Jaganathan and Runkle (1993) proposed GJR-GARCH model and Zakoian (1994) developed the Threshold GARCH (TGARCH) model. Lastly, Engle and Ng (1993) first proposed the Quadratic GARCH (QGARCH) model. The volatility is defined as the degree at which stock values or share price changes in the stock market.

There are also many researches in introducing heavy-tailed distributions into the GARCH framework to account for conditional heavy tails (leptokurtosis). For instance, Bollerslev (1987) introduce the Student's *t* distribution and the GARCH model with the Student's *t* distribution could capture dynamics of a variety of foreign exchange rates and stock price indices returns. Politis (2004) introduced the truncated standard normal distribution into the ARCH model and showed the empirical performance of the new type of heavy-tailed distribution on stock return. Tavares, Curto and Tavare (2007) model the heavy tails and asymmetric effect on stocks returns volatility into the GARCH framework, and showed the Student's t and the stable Paretian with ($\alpha < 2$) distribution clearly outperform the Gaussian distribution in fitting S&P 500 returns

As pointed out by Cont (2001), two of the eleven stylized facts for asset returns are: volatility clustering and conditional heavy tails. The volatility clustering says that different measures of volatility display a positive autocorrelation over several days, which quantifies the fact that high-volatility events tend to cluster in time; and the conditional heavy tails say that even after correcting returns for volatility clustering (e.g. via GARCH-type models), the residual time series still exhibit heavy tails. However, the tails are less heavy than in the unconditional distribution of returns.

The Mongolian Stock Exchange (MSE) was established in 1991 to implement the privatization of state-owned enterprises in the early stages of transition. With the enactment of the Securities and exchange Law (1994) and the Corporate Law (1995), a secondary market was established and brokerage firms operated and financed by th MSE were privatized. The MSE has seen rapid growth since 2006, when the economy enjoyed high economic growth. In 2010, the MSE was the world's best-performing stock market after an increase of 121 percent. In april 2011, the government signed the "Master Service Agreement" with the London Stock

Exchange, aiming to launch a modern trading system in Mongolia, to improve the legal environment, financial infrastructure and technology and to help the capacity building of trading staff. Within the scope of this agreement, the "Millennium IT" operating system was introduced to the MSE in 2012.The system is the fastest cash trading platform in the world and enables the stock exchange to provide substantially lower latency, significantly higher capacity and improved scalability to meet investors, professional brokers and dealer's needs.

There are few researchers particularly interested in the Mongolian stock markets. One study to note was done by V.Danaasuren (2010) when she tried to explain the stock price volatility using changes in fundamentals. One of the findings of the study was that, although stock market index explained a large portion of the stock returns variability, its effect or influence was weak in comparison to changes in economic variables. Hence, the conclusion was that stock returns are determined by economic factors. V.Danaasuren (2012) also provides an analysis of high-frequency recordings of the Mongolian share price index. She also used to GARCH type of models. This study considered– the leverage effect – the asymmetric impact of positive and negative information on the level of future variance.

The aim of this paper is to introduce several volatility models and use these models to predict the conditional variance about the return in different stock markets. I use the GARCH model, EGARCH model and GJR-GARCH model to analyze the return and consider using 2 different distributions on the error term: normal distribution and student-t distribution. In addition, this paper is mainly capturing the forecasting performance in the stock market with volatility models under different error distributions. Finally, after comparing the Root Mean Square Error (RMSE), choose the best model to predict the conditional variance. If the distribution or the variability of returns is known, then it would be possible to forecast returns with higher accuracy.

The paper is organized as follows: Section 2 describes the the models used, Section 3 discusses the data description and preliminary analysis, section 4 is about result of estimation of models, Section 5 is about empirical result and section 6 concludes the paper.

2. Model

This paper chooses GARCH model, E-GARCH model and GJR-GARCH model to analyze the stock returns. I selected the 3 stock market indexes: Standard and Poor's 500 daily index, FTSE100 daily index and MSE (Mongolian stock exchange) daily index.

2.1 ARCH model

Engle (1982) proposed the ARCH model (Auto-regressive Conditional Heteroskedastic Model).

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \ldots + \alpha_q \varepsilon_{t-q}^2 \tag{1}$$

Baillie and Bollerslev (1989) explained the variation on error terms has been changed from the constant to be a random sequence. Teräsvirta (2006) pointed out, ε_t has a conditional mean and variance based on the information set I_{t-1}.

$$\mathrm{E}(\varepsilon_t | I_{t-1}) = 0$$

$$\sigma_t^2 = \mathrm{E}(\varepsilon_t | I_{t-1}) \tag{2}$$

Here,

$$\varepsilon_t = z_t \sigma_t$$

$$z_t \sim N(0,1),$$

So $\{\epsilon_t\}$ is a normal distribution which mean equals to zero and variance equals to σ_t^2 ,

$$\varepsilon_t \sim N(0, \sigma_t^2)$$

Assume $\alpha_0 > 0$, and $\alpha_t > 0$, t = 1, . . . , q, $\alpha_1 + \ldots + \alpha_q < 1$ for ensuring $\{\sigma_t^2\}$ as weak stationary.

2.2 Generalized-ARCH model (GARCH)

Robert Engle (1982) proposed the Autoregressive Conditional Heteroscedasticity (ARCH) model which states that the variance in the data at time t depends on the previous time t-1. Tim Bollerslev (1986) generalized the ARCH model and named the model as the Generalized ARCH (GARCH) model which allows for a more flexible lag structure and it bears much resemblance to the extension of the standard times series autoregressive (AR) process to the general autoregressive moving average (ARMA) process.

Bollerslev proposed the GARCH (p, q) model given by:

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2$$

(3)

Bollerslev (1986) and Taylor (1986) assume $\alpha_0 > 0$; $\alpha_i \geq 0$, i = 1, . . . , q; $\beta_j \geq 0$, j = 1, . . . , p; $\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j < 1$ for ensuring σ_t^2 as weak stationary. GARCH is most used model for volatility however he has some limitations on GARCH model. Such models cannot take into consideration asymmetry, leverage effects, and coefficient restrictions. The most important limitation of GARCH model cannot capture the asymmetric performance. Later, for improving this problem, Nelson (1991) proposed the EGARCH model and Glosten, Jagannathan and Runkel (1993) proposed GJR-GARCH model.

The probability distribution of asset returns often exhibits fatter tails than the standard normal distribution. The existence of heavy-tailed is probably due to a volatility clustering in stock markets. In addition, another source for heavy-tail seems to be the sudden changes in stock returns. An excess kurtosis also might be originated from fat tailed. Moreover, in practice, the returns are typically negatively skewed. In order to capture this phenomenon (e.g., heavy-tailed), the GED (Generalized error distribution) is also considered in our analysis.[1]

1 Walter Enders – Applied econometric time series

2.3 Exponential GARCH (EGARCH) model

Nelson (1991) proposed the exponential GARCH (EGARCH) model.

$$\log\sigma_t^2 = c + \sum_{i=1}^{p} g(Z_{t-i}) + \sum_{j=1}^{q} \beta_j \log \sigma_{t-j}^2 \qquad (4)$$

Where,

$$g(Z_{t-i}) = \gamma_i Z_{t-i} + \alpha_i(|Z_{t-i}| - E(|Z_{t-i}|) \qquad (5)$$

Define $Z_{t-i} = \frac{\varepsilon_{t-i}}{\sigma_{t-i}}$ and the nature logarithm of the conditional variance equals to:

$$\log\sigma_t^2 = c + \sum_{j=1}^{q} \beta_j log\sigma_{t-j}^2 + \sum_{i=1}^{p} \alpha_i \left(\left|\frac{\varepsilon_{t-i}}{\sigma_{t-i}}\right| - E\left(\left|\frac{\varepsilon_{t-i}}{\sigma_{t-i}}\right|\right)\right) + \sum_{i=1}^{p} \gamma_i \frac{\varepsilon_{t-i}}{\sigma_{t-i}} \qquad (6)$$

In equation (6), α represents the asymmetric effect, β measures the lagged conditional variances and γ reflect the asymmetric performances.

There are 3 interesting features to notice about the EGARCH model:

1. The equation for the conditional variance is in log-linear form. Regardless of the magnitude of $\log\sigma_t^2$, the implied value of σ_t^2 never be negative.

2. Instead of using the value of ε_{t-1}^2, the EGARCH model uses the level of standardized value of ε_{t-i}. Nelson argues that this standardized allows for a more natural interpretation of the size and persistence of shocks. After all, the standardized value of ε_{t-i} is a unit free measure.

3. The EGARCH model allows for leverage effects.

The probability distribution:

$$\boxed{E(|Z_{t-i}|)} = \begin{cases} \sqrt{\frac{\pi}{2}} & \text{When } Z_{t-i} \text{ is normal distribution} \\ \frac{\sqrt{v}\,\Gamma[0.5(v-1)]}{[\sqrt{\pi}\,\Gamma(0.5v)]} & \text{When } Z_{t-i} \text{ is student- t distribution} \end{cases}$$

Wang, Fawson, Barrett and Mcdonald (2001) demonstrate $E(|Z_{t-i}|)$ is constant for all i when z_t is normal distribution or is $\frac{\sqrt{v}\Gamma[0.5(v-1)]}{[\sqrt{\pi}\Gamma(0.5v)]}$ depended on different v When Z_t is student- t distribution.

2.4 GJR-GARCH model

Glosten, Jagannathan and Runkle (1993) proposed GJR-GARCH model as another asymmetric model. Define the sequence $\{e_t\}$ equals to $z_t\,\sigma_t$ and e_t is a normal distribution.

$$\varepsilon_t \sim N(0, \sigma_t^2)$$

So the GJR-GARCH model is written by:

$$\sigma_t^2 = c + \sum_{j=1}^{p} \beta_j\, \sigma_{t-j}^2 + \sum_{i=1}^{p} \alpha_i\, \varepsilon_{t-i}^2 + \sum_{k=1}^{r} \gamma_k \varepsilon_{t-k}^2 I_{t-k}(\varepsilon_{t-k} < 0) \tag{8}$$

In GJR-GARCH model, the sign of the indicator term captures the asymmetry and Patrick, Stewart and Chris (2006) describes it in details in their article.

$$I_t = \begin{cases} 1 & if\ \varepsilon_t < 0 \\ 0 & otherwise \end{cases}$$

Where I_t is an indicator function, when the residual (ε_t) is smaller than zero, the indicator term (I_t) equals to one or equals to zero when the residual is not smaller than zero.

2.5 Distribution of the error term

This paper mainly introduces two distributions. One is normal distribution, the other one is Student -t distribution.

2.5.1 Normal distribution

The normal distribution is the most important and most widely used distribution in statistics. Normal distributions can differ in their means and in their standard deviations. Figure 1 shows three normal distributions. The green (left-most) distribution has a mean of -3 and a standard deviation of 0.5, the distribution in red (the middle distribution) has a mean of 0 and a standard deviation of 1, and the distribution in black (right-most) has a mean of 2 and a standard deviation of 3. These as well as all other normal distributions are symmetric with relatively more values at the center of the distribution and relatively few in the tails.

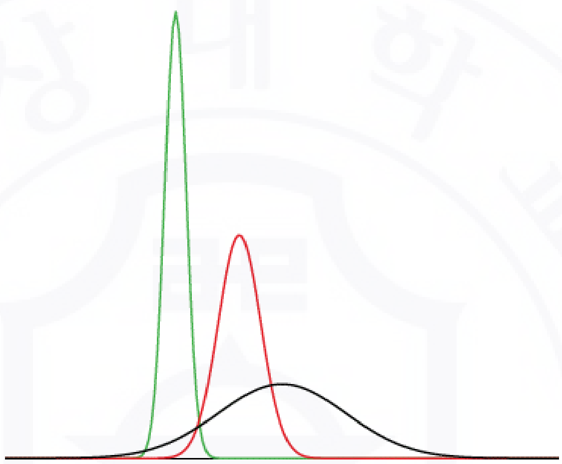

Figure 1. Normal distributions differing in mean and standard deviation.

The probability density function of Z_t is given as follows

$$f(z_t) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left\{-\frac{1}{2}\left(\left(\frac{z_t-\mu}{\sigma}\right)^2\right\}\right. \qquad (10)$$

The parameters μ and σ are the mean and standard deviation, respectively, and define the normal distribution. The symbol *e* is the base of the natural logarithm and π is the constant pi.

2.5.2 Student t-distribution

A statistical distribution published by William Gosset who published under the pseudonym *Student* in 1908. In particular, this distribution will arise in the study of a standardized version of the sample mean when the underlying distribution is normal. Conditional distribution of T is given:

$$T = \frac{Z}{\sqrt{V/n}} \tag{11}$$

The distribution of T is known as the Student t distribution with n degree of freedom. The distribution is well defined for any $n > 0$, but in practice, only positive integer values of n are of interest. Suppose that Z has the standard normal distribution, V has the chi-squared distribution with n degrees of freedom, and that Z and V are independent.

You will show that T has probability density function given by:

$$f(z_t) = \frac{\Gamma(\frac{v+1}{2})}{\Gamma(\frac{v}{2})\sqrt{(v-2)\pi}}\,(1 + \frac{z_t^2}{v-2})^{-\frac{1}{2}(v+1)} \tag{12}$$

Where v is the number of degree of freedom, $2 < v \leq \infty$, and Γ is gamma function. When $v \rightarrow \infty$, the student -t distribution nearly equals to the normal distribution. The lower the v, the fatter the tails.

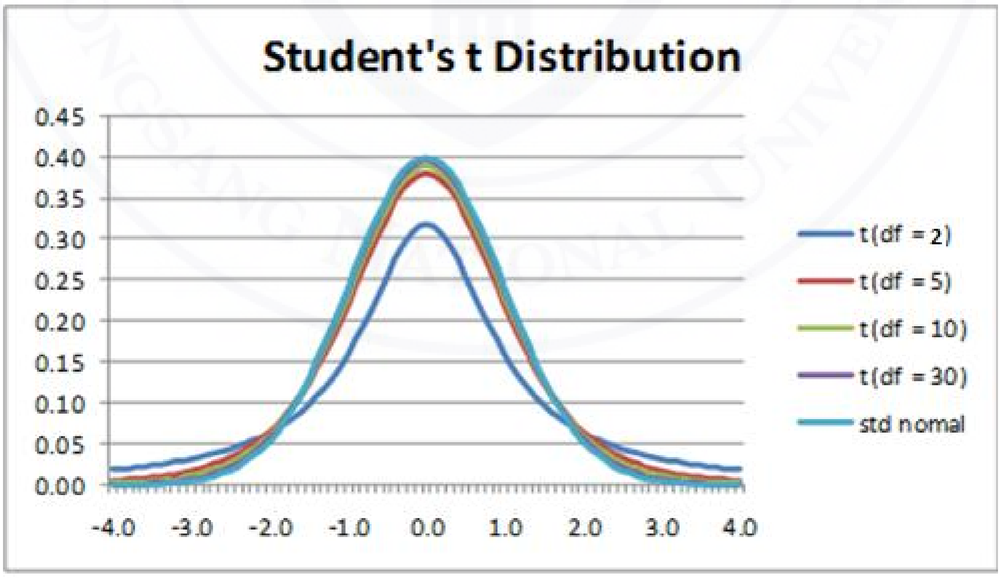

Figure 2. Student-t distribution with degree of freedom

2.6 Root Mean Square Error (RMSE)

Root Mean Square Error (RMSE) measures the difference between the true values and estimated values, and accumulates all these difference together as a standard for the predictive ability of a model. The criterion is the smaller value of the RMSE, the better the predicting ability of the model. This article uses this method to determine which model has the best forecasting performance.

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(r_t - \hat{r_t})^2}{n}} \qquad (13)$$

Where r_t is observed values and $\hat{r_t}$ is the predicted value of conditional variance at time i, n is the number of forecasts.

3. Data description and preliminary analysis

3.1 Data description

The data used in this study consist of daily Standard and Poor's 500 daily indexes, FTSE100 daily index and MSE daily index. MSE daily index is Mongolian stock market's top 20 company's data. The motivation, I chose MSE is there are few papers using MSE stock indexes. Moreover, there are no papers forecasting volatility for asymmetry with fat tailed distribution.

3.1.1 Sample period

Table 1. Sample periods of 3 stock indexes

Data	Sample period
S&P500	2005.12.31 – 2018.09.30
FTSE100	2007.12.31 – 2018.07.30
MSE	2012.01.01 – 2018.10.23

Table 1 below shows the time period of the data for each market.

3.1.2 Daily index

Figure 3 presents the daily indexes of the three market’s series during the considered time period.

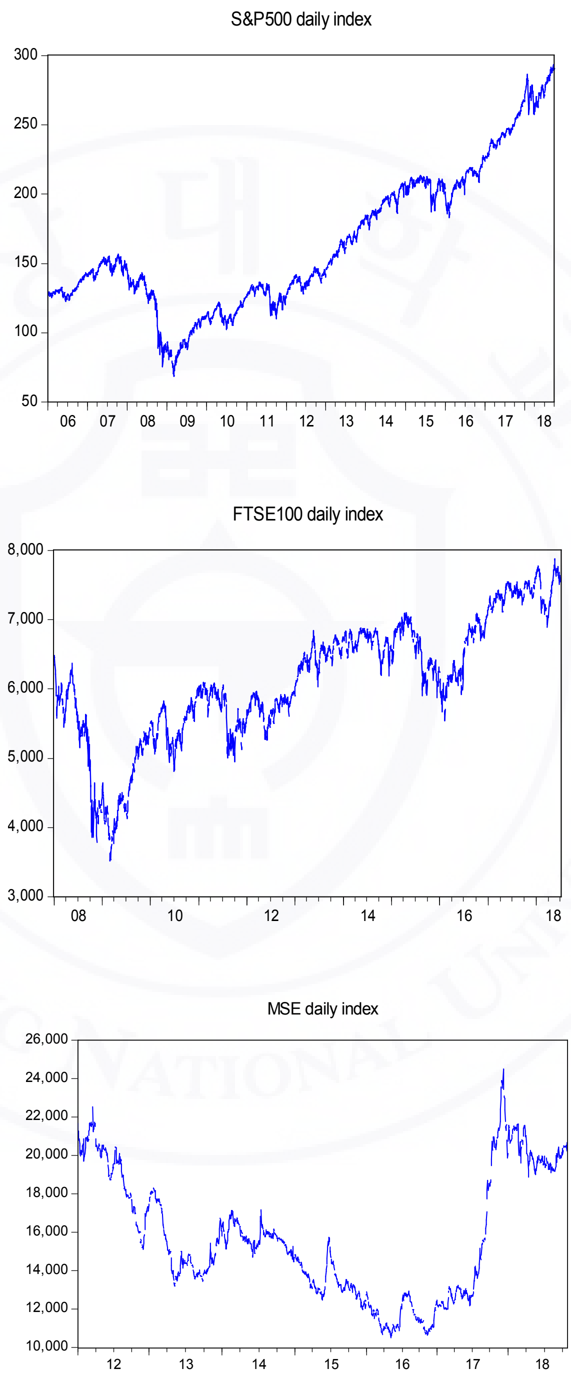

Figure 3. Three stock market’s daily index

3.1.3 Daily returns

Figure 4 presents the daily returns of the three market's series during the considered time period.

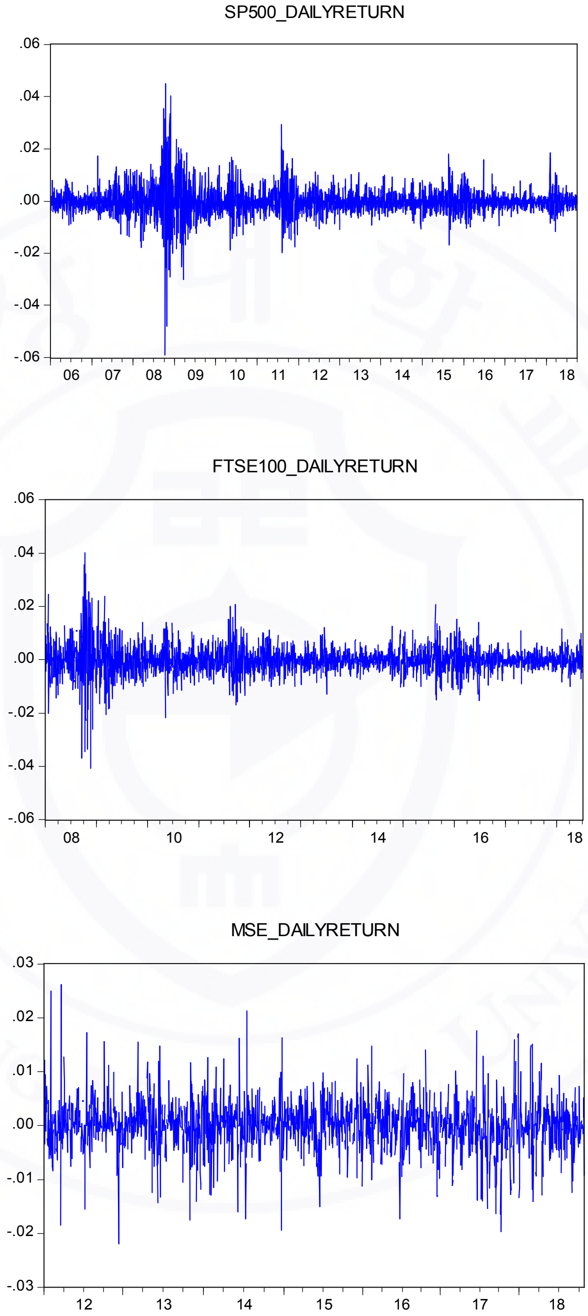

Figure 4. Three stock market's daily return

Table2. Descriptive statistics of 3 stock returns

	S&P500	FTSE100	MSE
Sample size	3208	2754	1706
Min	-0.058880	-0.040756	-0.26150
Max	0.045009	0.040240	0.022008
Mean	-0.00115	-2.58e-05	1.23e-05
SD	0.140560	0.005163	0.004733
Skewness	-0.2449	0.127936	0.095898
Kurtosis	17.85997	11.68828	6.312333
Jargue-Bera	9526.9660	8666.412	782.0507

Table 2 presents the summary statistics (mean, standard deviations, skewness, kurtosis, Jarque-Bera test) for daily stock returns.

From this table, the skewness are -0.2449, 0.127936, and 0.095898. These are not zero which means all of the returns are not symmetric. The kurtosis for different returns are 17.85997, 11.68828, and 6.312333. They are larger than three, which means all stock returns have the fat-tail characteristic.

It is widely known that the distributions of financial assets exhibit fatter tails than the normal distribution. In addition, the returns are often characterized by a number of stylized “facts” such as volatility clustering and volatility asymmetry.

3.2 Test for ARCH effect

As it was shown in the data description part when the residuals were examined for heteroscedasticity, ARCH-LM test provides strong evidence of ARCH effects in the residual series, which indicates that we can now proceed with the modelling of the index return volatility by using GARCH methodology.

Table3. Checklist of ARCH effect

Box-Ljung test		S&P500	FTSE100	MSE
	Test value	20.4826	2.467	9.100056
	P value	0.0000	0.0116	0.0000

The table 3 represents the result of ARCH effect of daily 3 market’s returns.

Based on the assumption of 5% significance level, all of the p-values are smaller than 0.05, which means the returns have ARCH effect.

3.3 Normality test

Table4.Normality test result

	S&P500	FTSE100	MSE
Jargue Bera	9526.9660	8666.412	782.0507
P-value	0.0000	0.0000	0.0000

The table 4 shows the result of normality test for daily S&P500, FTSE100, MSE's indexes

According to the normality test all 3 markets indexes probability is less than 5% which means we rejected the null hypothesis. Our data are not normality.

3.4 Summary of preliminary analysis

The daily indexes and returns for S&P500, FTSE100 and MSE index for the period under the review are presented in Figure 3 and Figure 4.

To specify the distributional properties of the daily S&P500, FTSE100, and MSE's returns series during the period of this study, various descriptive statistics were calculated and reported in Table 2. As can be seen from Table 2: The skewness of S&P500 return series are not close to zero which means the returns are not symmetric.

Likewise, in table4, the Jarque-Bera (J-B) statistic, which is a test for normality, also represents the non-normality for the daily S&P500, FTSE100, and MSE's returns.

One of the most important issues before applying the GARCH methodology is to first examine the residuals for evidence of heteroscedasticity. To test for the presence of heteroscedasticity in residuals of all index return series, the Lagrange Multiplier (LM) test for ARCH is applied. According to the result of LM-ARCH effect, we found there is a strong evidence for ARCH effect of three return series.

4. Estimation result

4.1 S&P500 and FTSE100 returns

This data is S&P500 daily return, from 2005-12-31 to 2018-09-30 in yahoo finance, a total of 3208 observations. This table points out all of the estimated parameters under different models.

Table5. Estimation result of S&P500 daily return

	GARCH		E-GARCH		GJR GARCH	
	Normal	**Student t**	**Normal**	**Student t**	**Normal**	**Student t**
C	0.000291	0.000355	-0.479064	-0.385032	4.74E-07	3.78E-07
(s.e)	5.89E-05	5.20E-05	0.31654	0.042702	4.14E-08	5.65E-08
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
α	4.13E-07	0.128964	-0.161580	-0.164043	-0.021270	-0.020266
(s.e)	4.38E-08	0.015110	0.011204	0.018752	0.007612	0.027204
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0052	0.0090
β	0.859171	0.870666	0.968102	0.977041	0.873388	0.870047
(s.e)	0.009451	0.013076	0.002514	0.003429	0.009046	0.012059
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ɣ			-0.176401	-0.183016	0.242675	0.264865
(s.e)			0.009312	0.015058	0.017587	0.027204
(p-value)			0.0000	0.0000	0.0000	0.0000
v		5.126511		5.685275		5.615951
(s.e)		0.519560		0.581321		0.605363
(p-value)		0.0000		0.0000		0.0000

Based on the assumption of 5% significance level, for GARCH (1, 1) model, when the error term is normal distribution or student-t distribution, all of the estimated parameters are significant. For EGARCH (1, 1) model, when the error term is normal distribution or student-t distribution all of the estimated parameters are significant. For GJR-GARCH (1, 1) model, whatever the error term is normal distribution or student-t distribution, all of the estimated parameters are significant.

This data is FTSE daily return, from 2007-12-31 to 2018-07-30 in yahoo finance, a total of 2754 observations. This table points out all of the estimated parameters under different models.

Table 6. Estimation result of FTSE100 daily return

	GARCH		E-GARCH		GJR GARCH	
	Normal	**Student t**	**Normal**	**Student t**	**Normal**	**Student t**
C	3.63E-07	2.96E-07	-0.297435	-0.291234	3.72E-07	3.51E-07
(s.e)	5.51E-08	8.10E-05	0.028037	0.039327	4.19E-08	5.96E-08
(p-value)	0.0000	0.0003	0.0000	0.0000	0.0000	0.0000
α	0.105409	0.112568	-0.118811	-0.130606	-0.011671	-0.016297
(s.e)	0.008819	0.014592	0.10952	0.017418	0.006989	0.011269
(p-value)	0.0000	0.0000	0.0000	0.0000	**0.0949**	**0.1481**
β	0.880171	0.881922	0.981020	0.982736	0.904763	0.89798
(s.e)	0.009587	0.013234	0.002188	0.003140	0.008049	0.011084
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ɣ			-0.119852	-0.142272	0.178619	0.206863
(s.e)			0.006856	0.012503	0.013496	0.022637
(p-value)			0000	0.0000	0.0000	0.0000
v		5.810073		6.680711		6.862077
(s.e)		0.789964		0.952911		1.036116
(p-value)		0.0000		0.0000		0.0000

Based on the assumption of 5% significance level, for GARCH (1, 1) model, whatever the error term is normal distribution or student-t distribution, all of the estimated parameters are significant. For EGARCH (1, 1) model, whatever the error term is normal distribution or student-t distribution, all of the estimated parameters are significant. For GJR-GARCH (1, 1) model, whatever the error term is normal distribution or student-t distribution, the estimated parameter α are not significant.

4.2 MSE daily returns

This data is MSE daily return, from 2012-01-01 to 2018-10-23 in yahoo finance, a total of 1705 observations. This table points out all of the estimated parameters under different models.

Table 7. Estimation result of MSE daily return

	GARCH		E-GARCH		GJR GARCH	
	Normal	Student t	Normal	Student t	Normal	Student t
C	5.51E-06	5.43E-06	-0.000205	-0.000240	5.57E-06	5.24E-06
(s.e)	5.70E-07	1.24E-06	9.85E-05	8.66E-05	5.76E-07	1.16E-06
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
α	0.198990	0.255653	-2.965820	-2.521411	0.278703	0.329499
(s.e)	0.021439	0.054575	0.292115	0.475653	0.032556	0.073677
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
β	0.561141	0.573168	0.747726	0.789609	0.563310	0.589817
(s.e)	0.034353	0.063816	0.026499	0.056712	0.034964	0.060993
(p-value)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
γ			0.116332	0.104471	-0.184522	-0.195873
(s.e)			0.018332	0.033758	0.034154	0.070488
(p-value)			0.0000	0.0020	0.0000	0.0055
v		3.535325		0.789608		3.594828
(s.e)		0.388813		0.043015		0.403784
(p-value)		0.0000		0.0000		0000

Based on the assumption of 5% significance level, for GARCH (1, 1) model, when the error term is normal distribution or student-t distribution, all of the estimated parameters are significant. For EGARCH (1, 1) model, when the error term is normal distribution or student-t distribution all of the estimated parameters are significant. For GJR-GARCH (1, 1) model, whatever the error term is normal distribution or student-t distribution, all of the estimated parameters are significant.

5. Forecasting performance

5.1 Out-of sample forecasting for S&P500 and FTSE100 returns

In order to acquire the appropriate model to forecast the conditional variance, this paper use out-of-sample to calculate the root mean square error (RMSE), and the detail can be got from Forsberg and Bollerslev, (2002).

Here, use S&P500 as an example to explain the process in details. The S&P500 stock return includes 3208 observations and reserve the last year as out-of-sample, including 252 observations. Put in-sample data into a window, so the length of window is fixed which equals to 1007. First, pick up the observations from 1 to 1007 into this fixed window and use GARCH models to estimate and forecast. In this way, I get the first prediction conditional variance. This process is called as one-step-ahead forecast. Second, repeat the first step except pick up the observations from 2 to 1008 in the fixed window, and get the second prediction conditional variance. Next, repeat the first step except pick up the observations from 3 to 1009 in the fixed window, and get the third prediction conditional variance. Repeat this step 252 times. We call such a process as multi-step-ahead forecast. Finally, use the 252 prediction conditional variance to calculate the RMSE by the formula (13).

With the same method, the FTSE100 daily return includes 2753 observations and reserve the last year as out-of-sample, including 251 observations. Put in-sample data into a window, so the length of window is fixed which equals to 1011. First, pick up the observations from 1 to 1011 into this fixed window and use GARCH models to estimate and forecast. In this way, I get the first prediction conditional variance. This process is called as one-step-ahead forecast. Second, repeat the first step except pick up the observations from 2 to 1012 in the fixed window, and get the second prediction conditional variance. Next, repeat the first step except pick up the observations from 3 to 1013 in the fixed window, and get the third prediction conditional variance. Repeat this step 251 times. We call such a process as multi-step-ahead forecast. Finally, use the 251 prediction conditional variance to calculate the RMSE by the formula (13)

Table8. The result of RMSE for S&P500 and FTSE100 returns

RMSE result		
S&P 500	GARCH (1,1)-Normal	0.52270
	GARCH (1,1)-Student t	0.52301
	EGARCH (1,1)-Normal	0.52260
	EGARCH (1,1)-Student t	**0.52241**
	GJR-GARCH (2,1)-Normal	0.52247
	GJR-GARCH (2,1)-Student t	0.52250
FTSE100	GARCH (1,1)-Normal	0.51642
	GARCH (1,1)-Student t	0.51642
	EGARCH (1,1)-Normal	0.51636
	EGARCH (1,1)-Student t	**0.51620**
	GJR-GARCH (1.1)-Normal	0.51626
	GJR-GARCH (1.1)-Student t	**0.51620**

The table 8 shows the result of RMSE of S&P500 and FTSE100 forecasting

Boldfaced word represents the minimal value of RMSE in each group. For each application, different models have different RMSE. For different applications, the model with minimal RMSE is also different.

For S&P500 stock return, E-GARCH (1, 1) model under student-t distribution has the smallest RMSE, so it will predict the future conditional variance better than other models.

For FTSE100 stock return, EGARCH and GJR-GARCH (1, 1) model under student-t distribution has the smallest RMSE, so it will forecast the future conditional variance better than other models. Our findings are consistent with the empirical evidence in previous studies such as Akbar and Nobi, (2018), Robert.D, Marta.S, (2015).

5.2 Out-of sample forecasting for MSE returns

With the same method, the MSE daily return includes 1706 observations and reserve the last year as out-of-sample, including 300 observations. Put in-sample data into a window, so the length of window is fixed which equals to 1011. First, pick up the observations from 1 to 1011 into this fixed window and use GARCH models to estimate and forecast. In this way, I get the first prediction conditional variance. This process is called as one-step-ahead forecast. Second, repeat the first step except pick up the observations from 2 to 1012 in the fixed window, and get the second prediction conditional variance. Next, repeat the first step except pick up the observations from 3 to 1013 in the fixed window, and get the third prediction conditional variance. Repeat this step 300 times. We call such a process as multi-step-ahead forecast. Finally, use the 300 prediction conditional variance to calculate the RMSE by the formula (13)

Table9. The result of RMSE for MSE return

RMSE result		
MSE	GARCH (1,1)-Normal	0.47390
	GARCH (1,1)-Student t	0.47380
	EGARCH (1,1)-Normal	0.47360
	EGARCH (1,1)-Student t	**0.47350**
	GJR-GARCH (1,1)-Normal	0.47370
	GJR-GARCH (1,1)-Student t	0.47380

The table 9 shows the result of RMSE for MSE forecasting

Boldfaced word represents the minimal value of RMSE for MSE. For MSE stock return, E-GARCH (1, 1) model under student-t distribution has the smallest RMSE, so it will predict the future conditional variance better than other models.

6. Conclusion

We find that most of the financial markets, characterized by feature of volatility, this property means a large and abnormal fluctuations. In, for instance, prices of shares and bonds. Naturally enough, these fluctuations are undesirable by investors and decisions- makers as they result in a state of uncertainty in financial transactions and thus, negatively affecting the economy. To deal with such problems, there is a need to use statistical models that take into account these fluctuations. One of these is the GARCH models.

This paper use different volatility models to analyze and forecast the conditional variance. At the same time, choose the normal distribution and the student-t distribution as the error term's distribution. Table 8, 9 illustrates which model has the smallest RMSE for different applications. Choose it as the best appropriate one to forecast the conditional variance. Different application has different appropriate model.

This article introduces two distributions from systems of frequency curves into the ARCH context and illustrates their usefulness in flexible modeling asymmetry and fat-tail of financial returns data. According to the stylized facts of the empirical distributions of returns, we have considered in this work 3 conditional heteroskedastic models, of which three (GARCH, EGARCH, and GJR-GARCH) allow the incorporation of volatility clusters, leverage effect, and partially, the heavy tails, when assuming that the conditional distribution of the assets returns is Normal and Student-t.

For S&P500 stock return, E-GARCH (1, 1) model under student-t distribution has the smallest RMSE, so it will predict the future conditional variance better than other models.

For FTSE100 stock return, EGARCH and GJR-GARCH (1, 1) model under student-t distribution has the smallest RMSE, so it will forecast the future conditional variance better than other models.

For MSE stock return, E-GARCH (1, 1) model under student-t distribution has the smallest RMSE, so it will predict the future conditional variance better than other models.

Asymmetric GARCH model (EGARCH, GJR-GARCH) with student-t distribution has better forecasting performance for S&P500, FTSE100, MSE stock returns.

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